Girish Chandar G

East Lansing, Michigan
+1(734)8810776 ganesang@msu.edu Homepage GoogleScholar

OBJECTIVE

Motivated Ph.D. student specializing in depth estimation, 3D reconstruction, and robot manipulation, with relevant research and industry experience working with foundation models, diffusion architectures, and VLAs.

EDUCATION

Michigan State University, MI, USA — **Jan 2024 – Present**
Ph.D. Computer Science, Advisor: Dr. Xiaoming Liu — *GPA : 4/4*

University of Michigan, Ann Arbor, MI, USA — **Aug 2021 – Apr 2023**
M.S. Electrical and Computer Engineering — *GPA : 4/4*

Indian Institute of Technology Gandhinagar, Gandhinagar, India — **July 2016 – Aug 2020**
B.Tech. Electrical Engineering (minor Computer Science) — *GPA : 8.98/10*

COURSEWORK / SKILLS

- Foundations for Computer Vision (A)
- Machine Learning (A)
- Matrix Methods for Machine Learning and Signal Processing(A+)
- 3D Computer Vision (A)
- Deep Learning for Computer Vision (A)
- PyTorch, Tensorflow, MXNet
- MATLAB, LabVIEW
- Numpy, OpenCV, Sklearn, Pandas
- C++

PUBLICATIONS

DiVER: Disentangled Visual Embeddings for Robot Manipulation — **In Progress**
Girish Chandar G, Yuliang Guo, Xiaoming Liu

DepthAgent: Towards Better Universal Depth Estimation via Sample-wise Expert Selection — **In Review, NeurIPS 2026**
Jie Zhu, Girish Chandar G, Xiaoming Liu

UniDAC: Universal Metric Depth Estimation for Any Camera — **CVPR 2026**
Girish Chandar G, Yuliang Guo, Liu Ren, Xiaoming Liu

Unleashing the Power of Chain-of-Prediction for Monocular 3D Object Detection — **CVPR 2026**
Zhihao Zhang, Abhinav Kumar, Girish Chandar G, Xiaoming Liu

RePLAy: Remove Projective LiDAR Depthmap Artifacts via Exploiting Epipolar Geometry — **ECCV 2024**
Shengjie Zhu, Girish Chandar G*, Abhinav Kumar, Xiaoming Liu*

POSITION

Research Assistant | Dr. Xiaoming Liu, Michigan State University — **June 2023 - Present**

Graduate Student Research Assistant | Dr. Stella Yu, University of Michigan-Ann Arbor — **Jan 2023 – Apr 2023**

Research Intern | Dr. Yang Zheng, NVIDIA — **May 2022 – August 2022**

INTERNSHIPS

Long-Range Lane Detection for Autonomous Driving | *Stack AV* | PyTorch — **May 2026 - Present**

Stereo Hazard Detection | *NVIDIA* | PyTorch — **May 2022 - Aug 2022**
- Implemented end-to-end deep learning model using custom UNet as the backbone for feature map extraction.
- Outperformed baseline by achieving **zero** false positives.

Auto Shape Detection in Machine Vision | *Zentron Labs* | Python (Numpy, OpenCV) — **Oct 2020 – Aug 2021**
- Implemented Arc Detection algorithm that gives accuracies of **100%** on simulated data and 80% on real data.
- Improved Line and Circle Detection accuracies from **65% to 90%**

Optimization based Inverse Rendering | *The University of Texas at Dallas* | PyTorch — **May 2019 – July 2019**
- Implemented algorithm for 3D face reconstruction from 2D images.
- 3D Morphable Model (3DMM) used as aprori mesh for efficient inverse rendering.

PROJECTS

SAR-NeRF | PyTorch — **Apr 2023**
- Research focused on modifying NeRF for 3D reconstruction of complex-valued radar data.

Small NeRF | PyTorch — **Apr 2022**
- Implemented a modified version of NeRF to reduce training time and computational cost.
- Experimented with multiple architectures to determine the best approximation of the original NeRF.

Co-Tuning for Transfer Learning on TACO Dataset | PyTorch — **Dec 2021**
- Implemented and verified the novel transfer algorithm proposed in "Co-tuning for Transfer Learning".
- **First team** to implement co-tuning on TACO (Trash Annotations in Context) dataset.